ACCT102 Management Accounting

Week 12

Differential Analysis

Adapted from © McGraw-Hill Education for ACCT102 (BT)

Learning Objective

Identify relevant and irrelevant costs and benefits in a decision.

Key Concept

1. **Define the alternatives – Understand the decision.**
2. **Distinguish what cost/benefit is relevant and irrelevant.**
   - Future costs and benefits that differ between alternatives are relevant.
   - Relevant costs and benefits should be considered.
   - Irrelevant costs and benefits should be ignored.
3. **Use relevant costs and benefits for evaluation.**

Costs and benefits that are relevant in one decision situation may not be relevant in another context.

Differential Cost and Revenue

- Future cost that differs between alternatives – Differential costs.
  - Incremental costs
  - Avoidable costs

Example:
You are deciding whether to work in your hometown, or a more lucrative neighboring city. Your travel and food costs in your hometown is $500 per month but will be $750 per month in the city.

You will need to rent a place for $1,000 if you work in the city. You can live in your parent's place for free if you work in your hometown.

- Future revenue that differs between alternatives – Differential revenue.

Opportunity Cost

- The potential benefit that is given up when one alternative is selected over another.
- Needs to be considered in decision making.

Sunk Cost

- **Sunk cost are IRREVELANT** – costs that has already been incurred.
- For example, cost of current equipment.

- Historical costs themselves are not relevant to decisions.
- Decisions often requires predictions. Historical costs may be the best available basis for prediction of future costs.

Irrelevant Costs and Benefits

- Future costs and benefits that **do not differ** between alternatives.
- **Common costs** are costs that are shared for all alternatives.
  - E.g. Joint costs

I'm deciding which Playstation game to buy to play.



COMMON COST

GTA

or

FIFA





Variable and Fixed Costs

- **Generally, variable costs are relevant costs.**
- However, there are exceptions when:
  - Conventional variable costs may not in fact vary in alternatives concerning a fixed volume of output.

    E.g. utilities in trade-off decisions (100 MHs of producing A vs B)
  - Special orders that may not involve incurring additional variable costs.

    E.g. Special order from the same customer – may not incur additional selling costs.

- **Conventional fixed cost may be affected by a particular decision, and therefore relevant.**
  - Fixed costs change with the levels of activity extend to a new level or capacity range (volume related)
  - Fixed costs change when extending it out of the relevant time horizon (time related)
- Unavoidable fixed costs are irrelevant.

Total vs Differential Cost Approaches

The management of a company is considering:

(1) New labor saving machine that rents for $3,000 per year

(2) Reduces direct labor cost per unit will fall by $3. $3 x 5,000 units = $15,000

TOTAL COST APPROACH

	Current Situation	Situation With New Machine	Differential Costs and Benefits
Sales (5,000 units @ $40 per unit)	$ 200,000	$ 200,000	-
Less variable expenses:			
COGS - DM (5,000 units @ $14 per unit)	70,000	70,000	-
COGS - DL (5,000 units @ $8 and $5 per unit)	40,000	25,000	15,000
COGS - VOH (5,000 units @ $2 per unit)	10,000	10,000	-
Total variable expenses	120,000	105,000	-
Contribution margin	80,000	95,000	15,000
Less fixed expense:			
Other	62,000	62,000	-
Rent on new machine	-	3,000	(3,000)
Total fixed expenses	62,000	65,000	(3,000)
Net operating income	$ 18,000	$ 30,000	12,000

Total vs Differential Cost Approaches

The management of a company is considering:

(1) New labor saving machine that rents for $3,000 per year

(2) Reduces direct labor cost per unit will fall by $3. $3 x 5,000 units = $15,000

DIFFERENTIAL COST APPROACH

Net Advantage to Renting the New Machine		
Decrease in direct labor costs (5,000 units @ $3 per unit)	$	15,000
Increase in fixed rental expenses		(3,000)
Impact on profits (Annual cost savings)	$	12,000

Total vs Differential Cost Approaches

Using the differential approach is desirable for two reasons:

- Only rarely will enough information be available to prepare detailed income statements for both alternatives.
- Mingling irrelevant costs with relevant costs may cause confusion and distract attention away from the information that is critical.

	Current Situation	Situation With New Machine	
Sales (5,000 units @ $40 per unit)	$ 200,000	$ 200,000	
Less variable expenses:			
COGS - DM (5,000 units @ $14 per unit)	???	???	No info
COGS - DL (5,000 units @ $8 and $5 per unit)	40,000	25,000	
COGS - VOH (5,000 units @ $2 per unit)	10,000	10,000	
Total variable expenses	50,000	35,000	
Contribution margin	???	???	Incomplete
Less fixed expense:			
Other (PARTIAL)	30,000	30,000	Partial
Rent on new machine	-	3,000	
Total fixed expenses	30,000	33,000	
Net operating income	???	???	Incomplete

Relevant Cost Analysis

1. **Define the alternatives – Understand the decision.**

2. **Distinguish what cost/benefit is relevant and irrelevant.**

   - Relevant costs and benefits should be considered.
   - Irrelevant costs and benefits should be ignored.

3. **Use relevant costs and benefits for evaluation.**

Illustration: Identifying Relevant Costs

Cynthia, a Malaysian student studying in Penang, is considering visiting her friend in Kuala Lumpur.

She can drive or take the budget airline. By car, it is 230 miles to her friend's apartment.

She is trying to decide which alternative is less expensive and has gathered the following information.

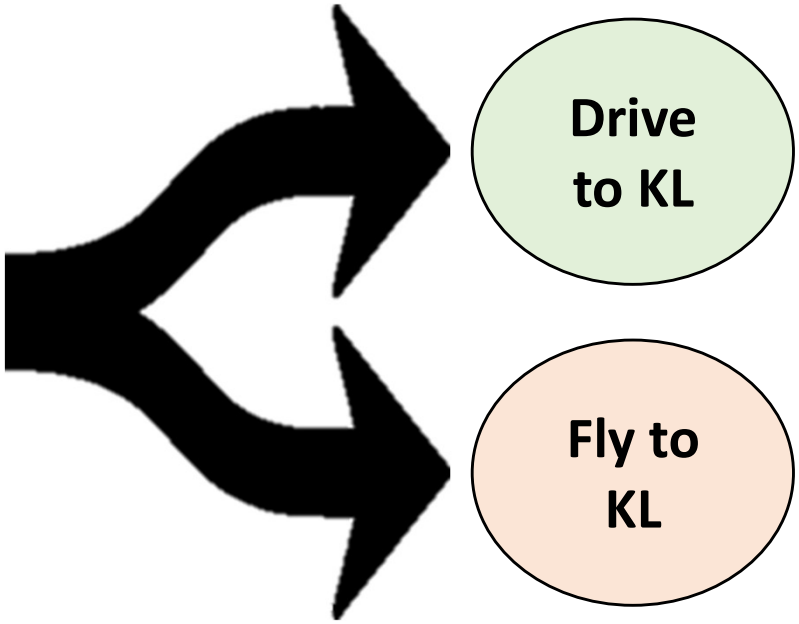


Illustration: Identifying Relevant Costs

Automobile Costs (based on 10,000 miles driven per year)			
$24,000 cost – $10,000 salvage ÷ 5 years		Annual Cost of Fixed Items	Cost per Mile
1	Annual straight-line depreciation on car	$ 2,800	$ 0.280
2	Cost of gasoline		0.100
3	Annual cost of auto insurance and license	1,380	0.138
4	Maintenance and repairs		0.065
5	Parking fees at school	360	0.036
	Total average cost		$ 0.619

Some Additional Information		
6	Reduction in resale value of car per mile of wear	$ 0.026
7	Round-tip airfare	$ 200
8	Benefits of relaxing on plane trip	????
9	Cost of putting dog in kennel while gone	$ 40
10	Benefit of having car in Kuala Lumpur	????
11	Hassle of parking car in Kuala Lumpur	????
12	Per day cost of parking car in Kuala Lumpur	$ 25

Adapted from © McGraw-Hill Education for ACCT102 (BT)

Illustration: Identifying Relevant Costs

Relevant Financial Cost of Driving	
Gasoline (460 @ $0.100 per mile)	$ 46.00
Maintenance (460 @ $0.065 per mile)	29.90
Reduction in resale (460 @ $0.026 per mile)	11.96
Parking in Kuala Lumpur (2 days @ $25 per day)	50.00
Total	$ 137.86

Relevant Financial Cost of Taking the Plane	
Round-trip ticket	$ 200.00

- **From a financial standpoint, Cynthia would be better off driving to visit her friend.**
- Some of the non-financial relevant factors may influence her final decision.
  - Relaxing on plane + No hassle of finding parking **vs** Benefit of having a car

Major Business Decisions

We shall be examining several decision-making processes next:

1. Addition/Dropping of product line or business segment or territory
2. Make or Buy (Outsourcing)
3. Accepting of special sales order
4. Sell or Process further of joint products

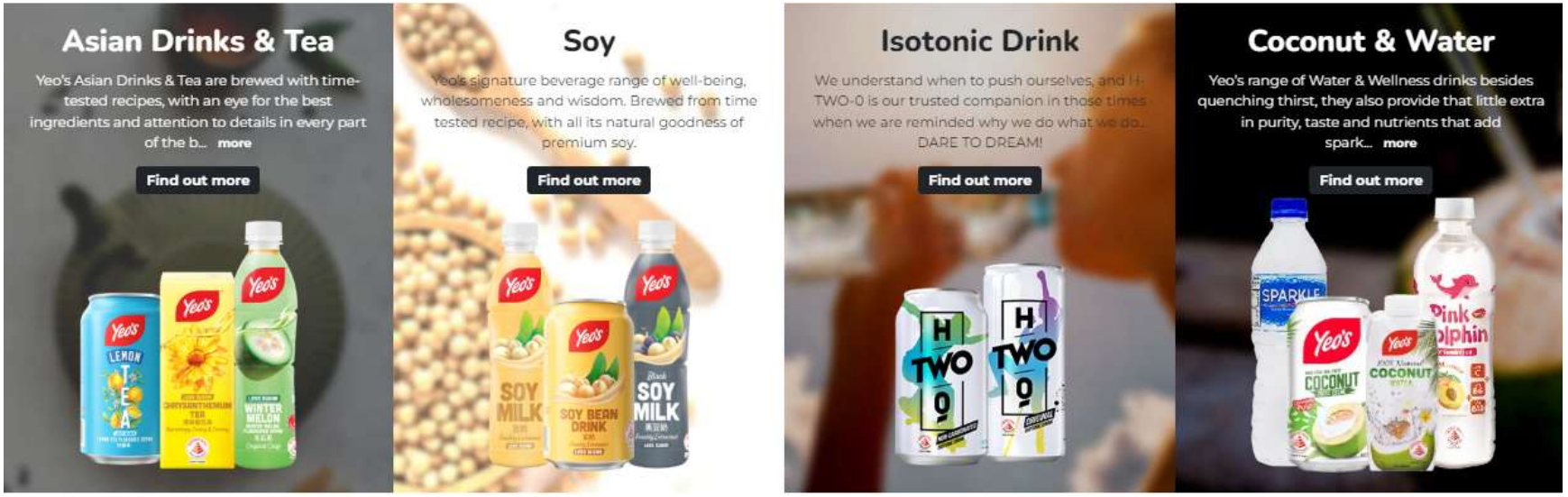
Learning Objective

Prepare an analysis showing whether a product line or other business segment should be dropped or retained.

Retain/Drop Segments

One of the most important decisions managers make is whether to retain or drop (or add) a business segment.

Ultimately, a decision to drop an old segment or add a new one is going to hinge primarily on the **impact on net operating income.**



Retain/Drop Segments

POINTS TO NOTE

- Relevant items are the change in revenue and variable costs.

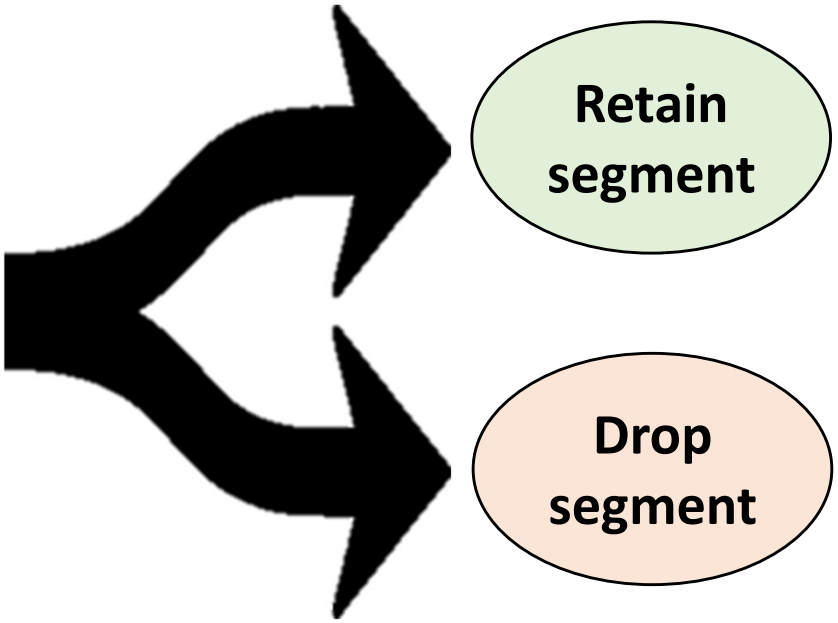
- Adding or dropping a business segment is mostly a long-run decision, therefore it is likely that some fixed costs will change and these would be relevant to the decision.

- If fixed costs do not change, fixed costs is irrelevant.
- If fixed costs change, those fixed costs become relevant.
  - Avoidable versus unavoidable
  - Incremental fixed costs

Retain/Drop Segments – Example

Due to the declining popularity of digital watches, Lovell Company's digital watch line has not reported a profit for several years.

Lovell is **considering discontinuing this product line.**



Contribution margin lost – Lower profits

vs

Cost avoided – Increase profits

DECISION RULE: Lovell should drop the digital watch segment only if its profit would increase.

Retain/Drop Segments – Example

After investigation, Lovell has provided some further information.

- **Only fixed general factory overhead and fixed general administrative expenses will not be affected by dropping the digital watch line.**
- Fixed general factory overhead and general administrative expenses assigned to this product would be reallocated to other product lines.
- **The equipment used to manufacture digital watches has no resale value or alternative use.**

Retain/Drop Segments – Comparative Income Approach

Comparative Income Approach Solution

	Retain	Drop	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:		-	
Variable COGS	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
General factory overhead	60,000		
Salary of line manager	90,000		
Depreciation	50,000		
Advertising - direct	100,000		
Rent - factory space	70,000		
General admin. expenses	30,000		
Total fixed expenses	400,000		
Net operating loss	$ (100,000)		

If the digital watch line is dropped, the company loses $300,000 in contribution margin.

Retain/Drop Segments – Comparative Income Approach

Comparative Income Approach Solution

	Retain	Drop	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:		-	
Variable COGS	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
General factory overhead	60,000	60,000	-
Salary of line manager	90,000		
Depreciation	50,000		
Advertising - direct	100,000		
Rent - factory space	70,000		
General admin. expenses	30,000		
Total fixed expenses	400,000		
Net operating loss	$ (100,000)		

General factory overhead reallocated and not affected by decision.

Retain/Drop Segments – Comparative Income Approach

Comparative Income Approach Solution

	Retain	Drop	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:		-	
Variable COGS	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
General factory overhead	60,000	60,000	-
Salary of line manager	90,000	-	90,000
Depreciation	50,000		
Advertising - direct	100,000		
Rent - factory space	70,000		
General admin. expenses	30,000		
Total fixed expenses	400,000		
Net operating loss	$ (100,000)		

The salary of the product line manager would be avoided.

Retain/Drop Segments – Comparative Income Approach

Comparative Income Approach Solution

	Retain	Drop	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:		-	
Variable COGS	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
General factory overhead	60,000	60,000	-
Salary of line manager	90,000	-	90,000
Depreciation	50,000	50,000	-
Advertising - direct	100,000		

The depreciation is a sunk cost. Also, remember that the equipment has no resale value or alternative use, so the equipment and the depreciation expense associated with it are irrelevant to the decision.

Retain/Drop Segments – Comparative Income Approach

Comparative Income Approach Solution

	Retain	Drop	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:		-	
Variable COGS	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
Gene		60,000	-
Sala		-	90,000
Depr		50,000	-
Advertising - direct	100,000	-	100,000
Rent - factory space	70,000	-	70,000
General admin. expenses	30,000	30,000	-
Total fixed exper		140,000	(260,000)
Net operating los		$ (140,000)	$ (40,000)

Advertising and rent expenses can be avoided if segment is dropped.

General admin expenses reallocated and not affected by decision.

Retain/Drop Segments – Comparative Income Approach

Comparative Income Approach Solution

	Retain	Drop	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:		-	
Variable COGS	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
General factory overhead	60,000	60,000	-
Salary of line manager	90,000	-	90,000
Depreciation	50,000	50,000	-
Advertising - direct	100,000	-	100,000
Rent - factory space	70,000	-	70,000
General admin. expenses	30,000	30,000	-
Total fixed expenses	400,000	140,000	(260,000)
Net operating loss	$ (100,000)	$ (140,000)	$ (40,000)

Calculate the differences between alternatives -> RETAIN

Retain/Drop Segments – Contribution Margin Approach

Digital Watches segment financials

Segment Income Statement
Digital Watches

Sales		$ 500,000
Less: variable expenses		
Variable COGS	$ 120,000	
Variable shipping costs	5,000	
Commissions	75,000	200,000
Contribution margin		$ 300,000
Less: fixed expenses		
~~General factory overhead~~	~~$ 60,000~~	
Salary of line manager	90,000	
~~Depreciation of equipment~~	~~50,000~~	
Advertising - direct	100,000	
Rent - factory space	70,000	
~~General admin. expenses~~	~~30,000~~	400,000
Net operating loss		$ (100,000)

Retain/Drop Segments – Contribution Margin Approach

Should Lovell drop the digital watch segment?

If Lovell drops the segment, net operating income will fall as the contribution margin lost is greater than the avoidable costs. Lovell should retain the segment.

Contribution Margin Solution

Contribution margin lost if digital watches are dropped		$ (300,000)
Less fixed costs that can be avoided		
Salary of the line manager	$ 90,000	
Advertising - direct	100,000	
Rent - factory space	70,000	260,000
Net disadvantage		$ (40,000)

Contribution margin lost > Cost avoided

Beware of Allocated Fixed Costs

- Beware of allocated fixed cost in keep or drop decisions.
- Fixed cost allocations can make a segment look less profitable than it really is.
- Including **unavoidable common fixed costs** makes the product line appear to be unprofitable.

Less: variable expenses		
Variable COGS	$ 120,000	
Variable shipping costs	5,000	
Commissions	75,000	200,000
Contribution margin		$ 300,000
Less: fixed expenses		
General factory overhead	$ 60,000	
Salary of line manager	90,000	$190,000
Depreciation of equipment	50,000	
Advertising - direct	100,000	
Rent - factory space	70,000	
General admin. expenses	30,000	400,000
Net operating loss		$ (100,000)

Quick Check 12.1

Snowy Ice Cream has 3 segments: Chocolate, Vanilla and Strawberry.

Segment financials (In $'000)	**Strawberry**
Sales	**700**
Variable Costs	490
Contribution Margin	**210**
Direct Fixed Costs	140
Allocated Fixed Costs	100
Net Income	**(30)**

20% of allocated fixed costs will be avoided if segment is dropped

If Snowy drops Strawberry segment, what will be the net advantage/disadvantage?

a) Net advantage of $35,000

b) Net disadvantage of $50,000

c) Net disadvantage of $210,000

d) Net disadvantage of $70,000

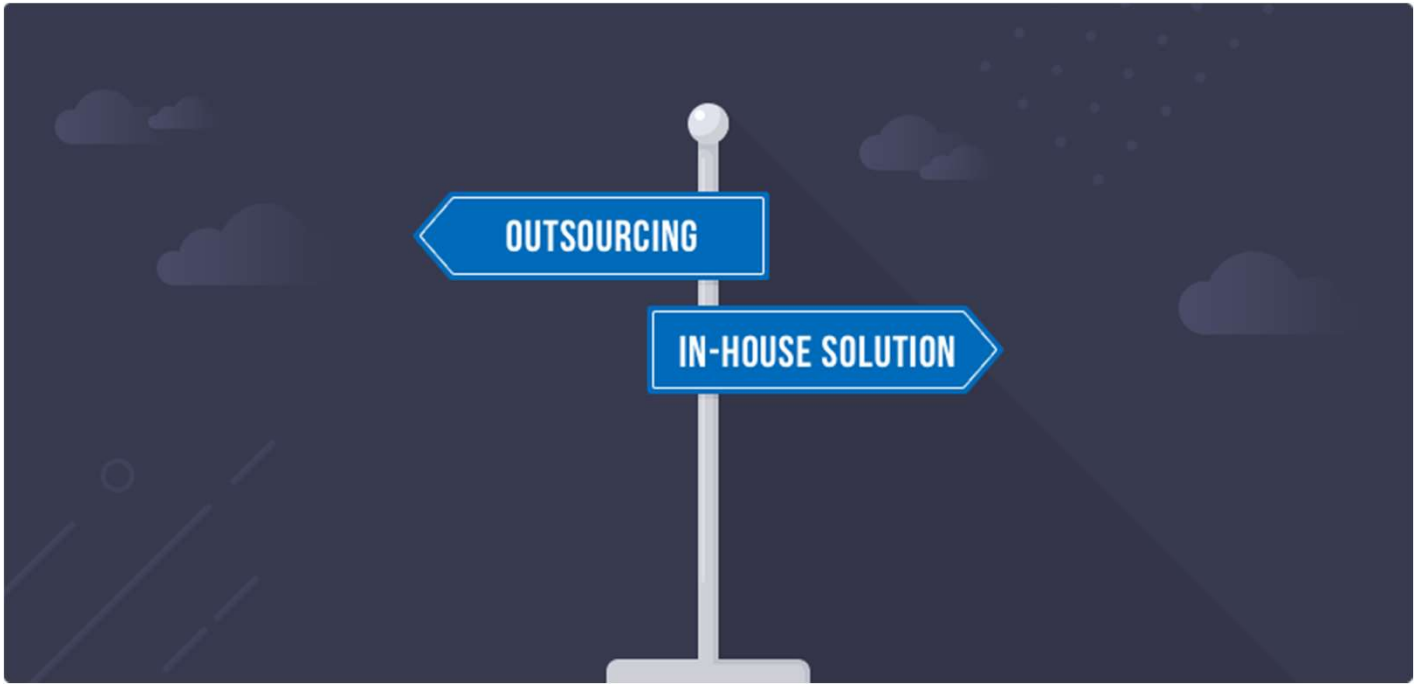
Learning Objective

Prepare a make or buy analysis.

The Make or Buy [Outsourcing] Decision

A decision to carry out one of the activities in the value chain internally, rather than to buy externally from a supplier is called a “make or buy” decision.

Qualitative factors often particularly important in outsourcing decisions.
E.g. Control over product quality, reliability of supply, product strategies



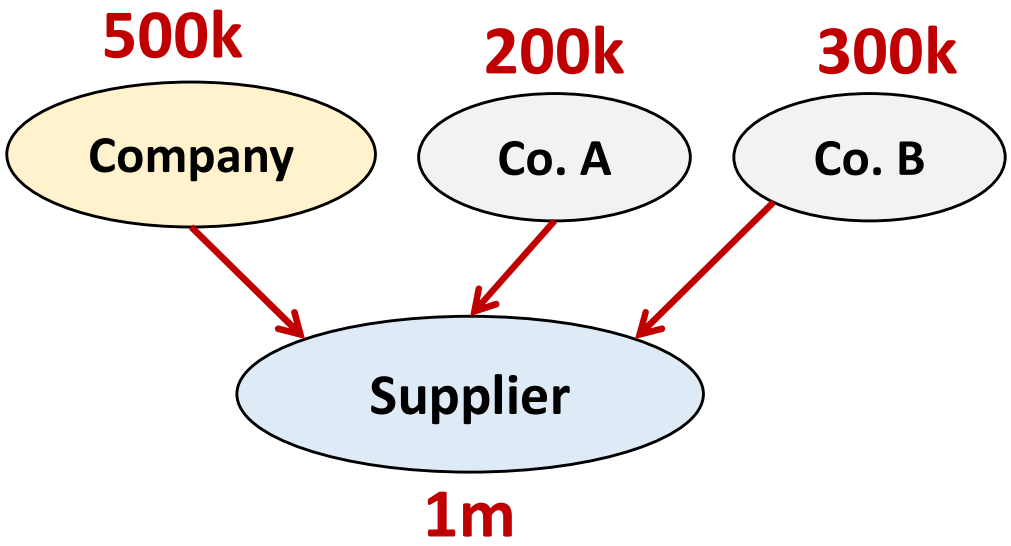
Vertical Integration

Advantages

- Smoother flow of parts and materials
- Better quality control
- Realize profits / cost savings

Disadvantages

- No economies of scale





Make or Buy – Example

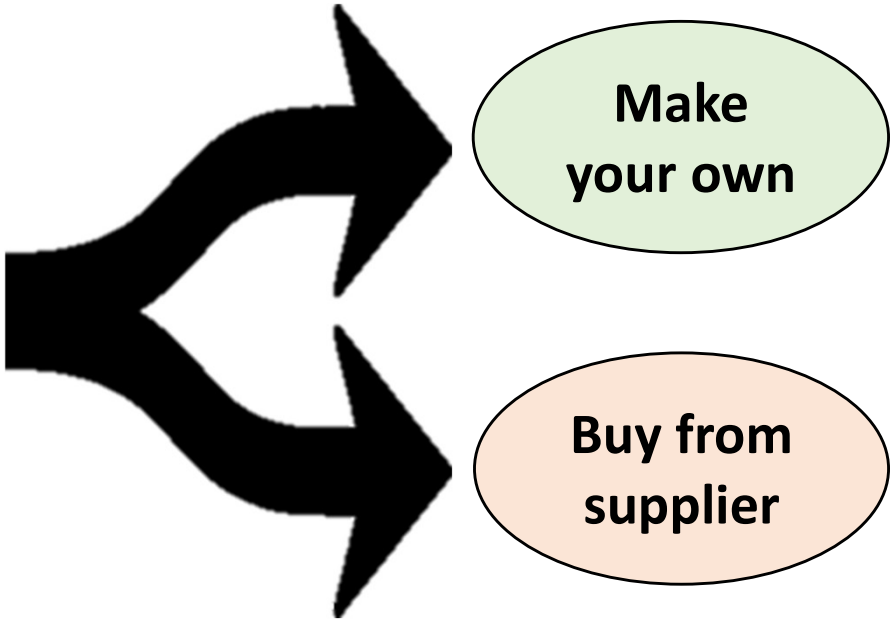
- Essex Company manufactures part 4A that is used in one of its products.
- The unit product cost of this part is:

Direct materials	$ 9
Direct labor	5
Variable overhead	1
Depreciation of special equip.	3
Supervisor's salary	2
General factory overhead	10
Unit product cost	$ 30

Make or Buy – Example

- The special equipment used to manufacture part 4A has no resale value.
- The total amount of general factory overhead, which is allocated on the basis of direct labor hours, would be unaffected by this decision.
- The $30 unit product cost is based on 20,000 parts produced each year.
- An outside supplier has offered to provide the 20,000 parts at a cost of $25 per part.

Should we accept the supplier's offer?



Make or Buy – Example

- **The special equipment used to manufacture part 4A has no resale value.**
- The total amount of general factory overhead, which is allocated on the basis of direct labor hours, would be unaffected by this decision.
- The $30 unit product cost is based on 20,000 parts produced each year.
- An outside supplier has offered to provide the 20,000 parts at a cost of $25 per part.

Should we accept the supplier's offer?

Unit product cost		
Direct materials	**$**	**9**
Direct labor		**5**
Variable overhead		**1**
~~**Depreciation of special equip.**~~		~~**3**~~
Supervisor's salary		**2**
General factory overhead		**10**
Unit product cost	**$**	**30**

The depreciation of the special equipment represents a **sunk cost**, thus **IRRELEVANT**. The equipment also has no resale value.

Make or Buy – Example

- The special equipment used to manufacture part 4A has no resale value.
- **The total amount of general factory overhead, which is allocated on the basis of direct labor hours, would be unaffected by this decision.**
- The $30 unit product cost is based on 20,000 parts produced each year.
- An outside supplier has offered to provide the 20,000 parts at a cost of $25 per part.

Should we accept the supplier's offer?

Unit product cost		
Direct materials	$	9
Direct labor		5
Variable overhead		1
~~Depreciation of special equip.~~		~~3~~
Supervisor's salary		2
~~General factory overhead~~		~~10~~
Unit product cost	$	30

If the product is dropped, general factory overheads are unaffected – probably **reallocated to other products**. Therefore, not avoidable and **IRRELEVANT**.

Make or Buy – Example

- The special equipment used to manufacture part 4A has no resale value.
- The total amount of general factory overhead, which is allocated on the basis of direct labor hours, would be unaffected by this decision.
- The $30 unit product cost is based on 20,000 parts produced each year.
- An outside supplier has offered to provide the 20,000 parts at a cost of $25 per part.

Should we accept the supplier's offer?

Unit product cost		
Direct materials	**$**	**9**
Direct labor		**5**
Variable overhead		**1**
~~**Depreciation of special equip.**~~		~~**3**~~
Supervisor's salary		**2**
~~**General factory overhead**~~		~~**10**~~
Unit product cost	**$**	**30**

The **avoidable costs** related with making part 4A include

- direct materials
- direct labor
- variable overhead, and
- the supervisor's salary

Therefore, **RELEVANT**

Make or Buy – Example

Should we make or buy part 4A?

Given that the total avoidable costs are less than the cost of buying the part, Essex should continue to make the part.

	Unit cost	Cost of 20,000 Units	
		Make	Buy
Supplier's price	$ 25		$ 500,000
Avoidable cost			
Direct materials	$ 9	180,000	
Direct labor	5	100,000	
Variable overhead	1	20,000	
Supervisor's salary	2	40,000	
	$ 17	$ 340,000	$ 500,000

Learning Objective

Prepare an analysis showing whether a special order should be accepted.

Special Orders

- A special order is a **one-time order** that is not considered part of the company's normal ongoing business.

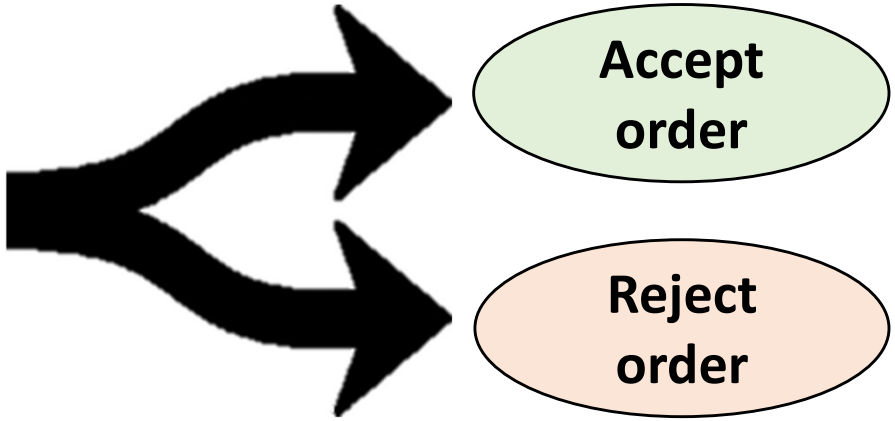
- When analyzing a special order, only the incremental costs and benefits are relevant.
- Since the existing fixed manufacturing overhead costs would not be affected by the order, they are irrelevant.

Special Orders – Example

- Jet Corp. makes a single product whose normal selling price is $20 per unit.
- A foreign distributor offers to purchase 3,000 units for $10 per unit.
- This is a one-time order that would not affect the company's regular business.
- Jet Corp. will still incur variable marketing costs on this special order.
- Annual capacity is 10,000 units, but Jet Corp. is currently producing and selling only 5,000 units.
- Jet Corp. determined that all fixed costs continue to be within the relevant range, even with the special order.

Jet Corp. has provided its variable costing income statement for you.

Should Jet accept the offer?



Special Orders – Example

Jet Corporation
Contribution Income Statement

Revenue (5,000 × $20)		$ 100,000
COGS (DM)	$ 20,000	
COGS (DL)	5,000	
COGS (VMOH)	10,000	
COGS		35,000
Var. Marketing costs		5,000
Total variable costs		40,000
Contribution margin		60,000
~~FMOH~~	~~$ 28,000~~	
~~Marketing costs~~	~~20,000~~	
Total fixed costs		48,000
Net operating income		$ 12,000

Special Orders – Example

Revenue (5,000 × $20)		$ 100,000
COGS (DM)	$ 20,000	
COGS (DL)	5,000	
COGS (VMOH)	10,000	
COGS		35,000
Var. Marketing costs		5,000
Total variable costs		40,000

$40k / 5k units = $8 variable cost per unit

Should Jet accept the offer?

Yes. If Jet accepts the special order, incremental revenue will exceed incremental costs. Net operating income will increase by $6,000.

Increase in revenue (3,000 × $10)	$30,000
Increase in costs (3,000 × $8 variable cost)	24,000
Increase in net income	$ 6,000

Learning Objective

Prepare an analysis showing whether joint products should be sold at the split-off point or processed further.

Joint Products

- In some industries, a number of end products are produced from a single raw material input.
- Two or more products produced from a common input are called **joint products**.
- The point in the manufacturing process where each joint product can be recognized as a separate product is called the **split-off point**.

Joint Products – Example

Petroleum refining industry

Intermediate Products

End Products

Joint costs are incurred up to the split-off point

Joint Products

Joint Input

Crude oil

Common Production Process

Oil → Additional Processing → Sale

Gasoline → Additional Processing → Sale

Chemicals → Additional Processing → Sale

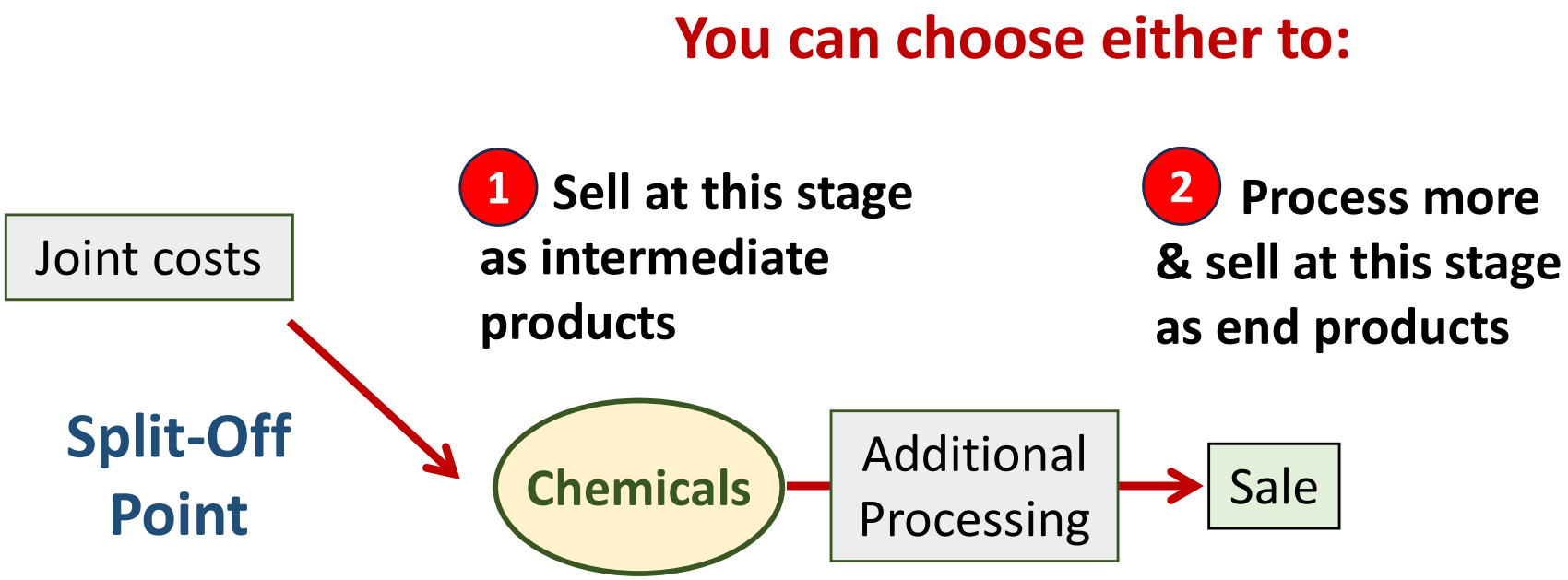


Split-Off Point

Incremental Processing Costs

Sell or Process Further

- With respect to sell or process further decisions, it is profitable to continue processing a joint product after the split-off point so long as **the incremental revenue from such processing exceeds the incremental processing costs incurred after the split-off point.**



Allocated Joint Costs

- Joint costs are **IRRELEVANT** in decisions regarding what to do with a product from the split-off point forward.
- Joint costs **should not** be considered sell or process further decisions

- Joint costs are often allocated as it is needed for some purposes such as balance sheet inventory valuation.
- Joint costs are traditionally allocated among different products at the split-off point.
- A typical approach is to allocate joint costs according to the relative sales value of the end products.
- Using allocated joint costs could lead to **wrong** decisions.

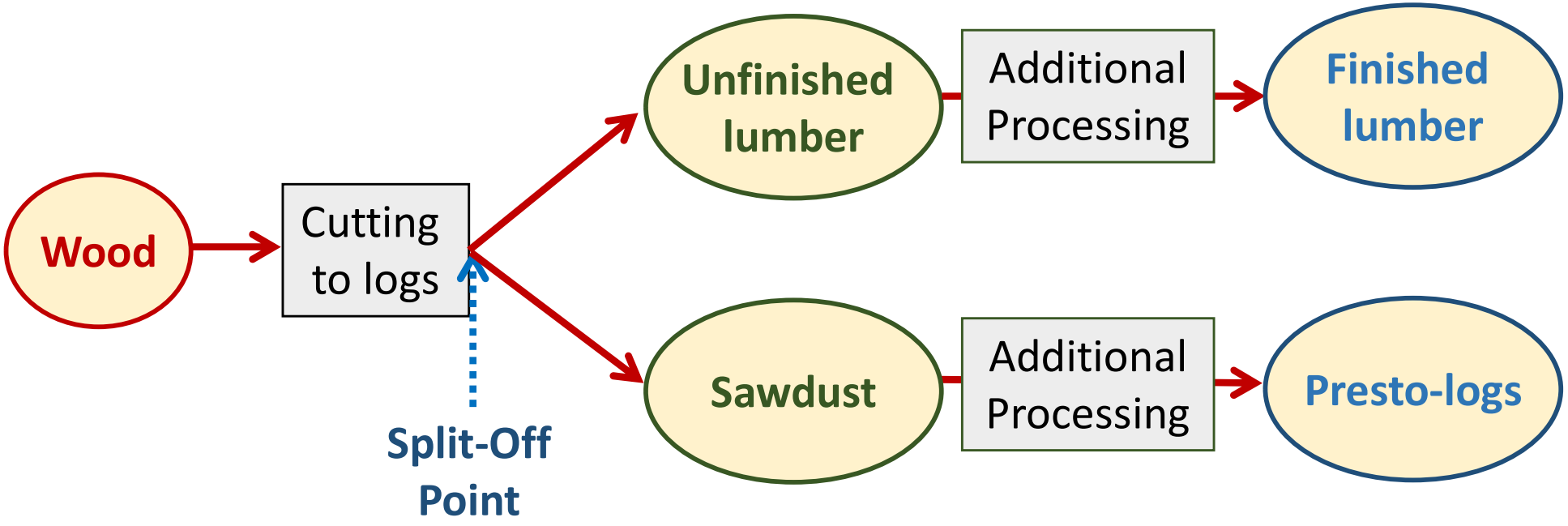
Sell or Process Further – Example

- Sawmill, Inc. cuts logs from which unfinished lumber and sawdust are the immediate joint products.
- Unfinished lumber is sold “as is” or processed further into finished lumber.
- Sawdust can also be sold “as is” to gardening wholesalers or processed further into “presto-logs.”



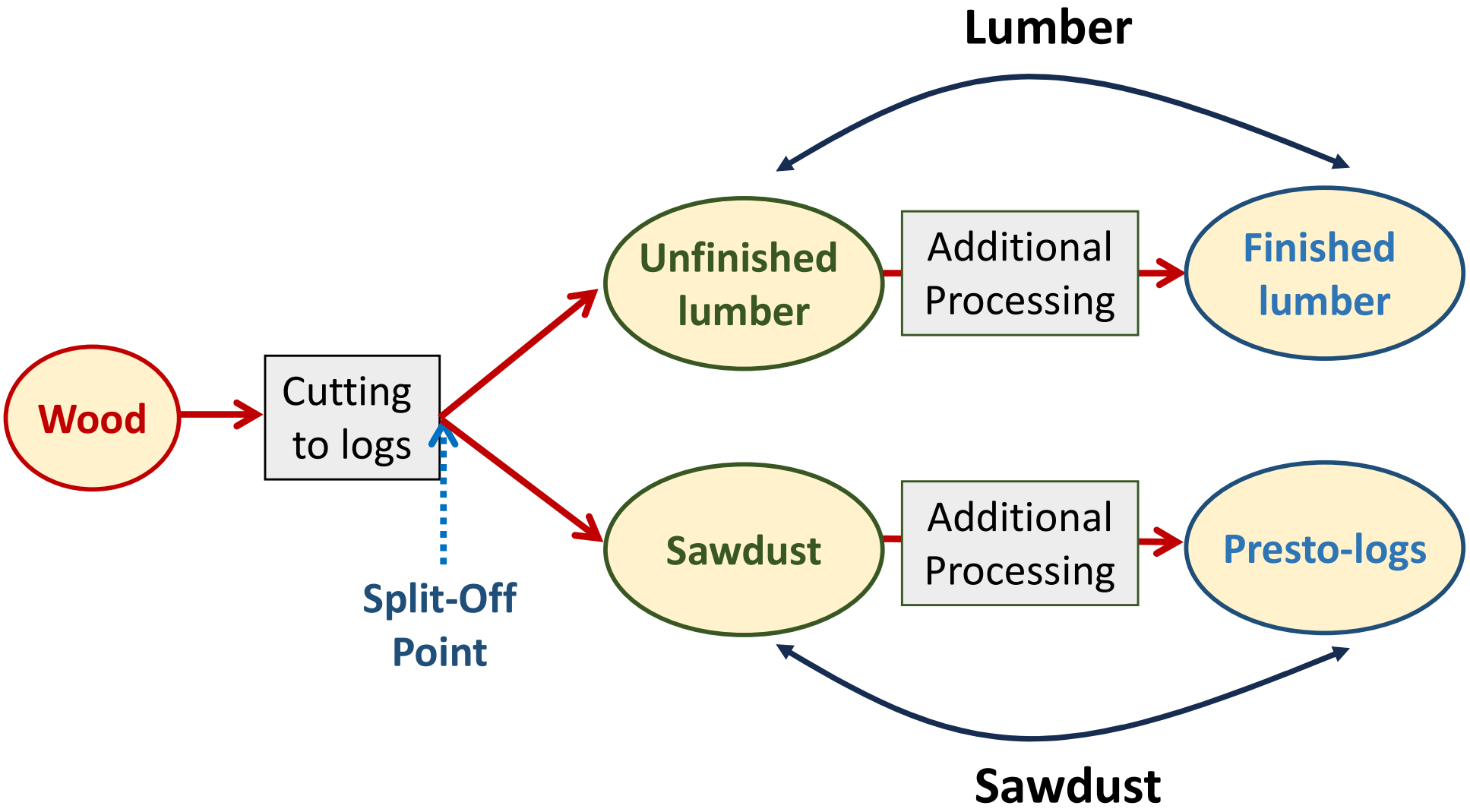
Sell or Process Further – Example

- Sawmill, Inc. cuts logs from which unfinished lumber and sawdust are the immediate joint products.
- Unfinished lumber is sold "as is" or processed further into finished lumber.
- Sawdust can also be sold "as is" to gardening wholesalers or processed further into "presto-logs."



Sell or Process Further – Example

- 2 sell or process further decisions to be made.



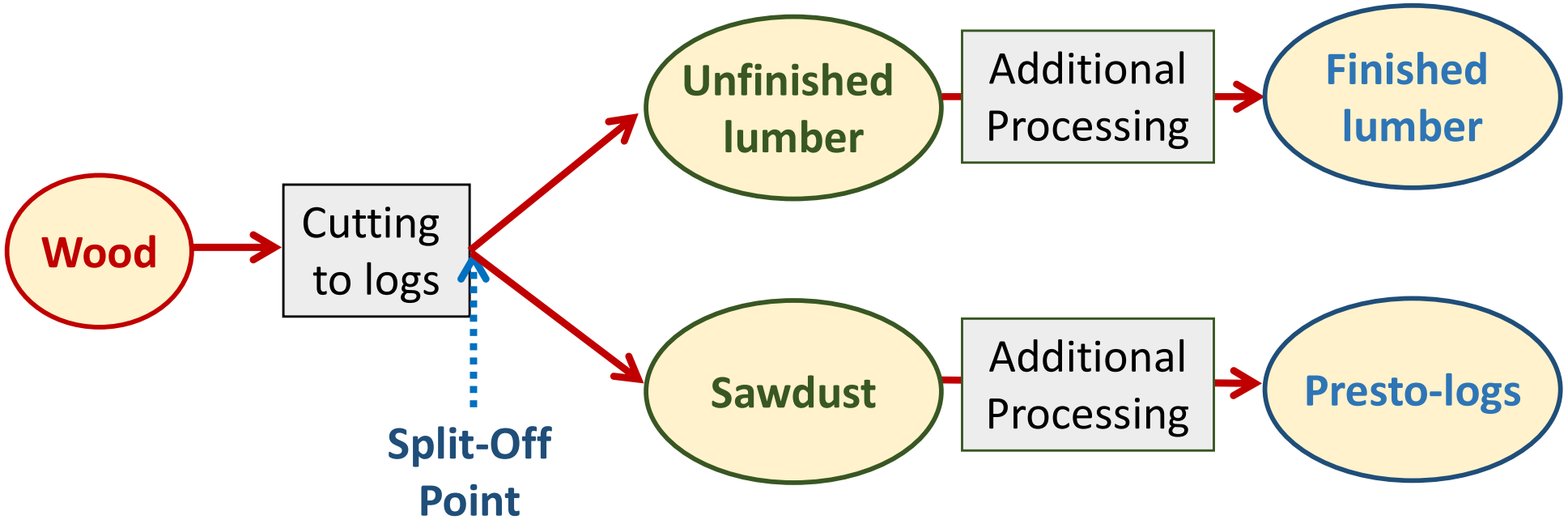
Sell or Process Further – Example

Data about Sawmill's joint products includes:

	$ Per Log	
	Lumber	Sawdust
Allocated joint product costs	176	24
Cost of further processing	50	20
Sales value at the split-off point	140	40
Sales value after further processing	270	50

Sell or Process Further – Example

	Allocated joint product costs	Sales value at spilt-off pt	Cost of add. processing	Sales value aft. add. processing
Lumber	**$176 allocated cost**	**Can sell for $140**	**Further cost Of $50**	**Can sell For $270**
Sawdust	**$24 allocated cost**	**Can sell for $40**	**Further cost Of $20**	**Can sell For $50**



Sell or Process Further – Example

What is relevant?

- Incremental revenue from such processing
- Incremental processing costs incurred after the split-off point.

	$ Per Log	
	Lumber	Sawdust
~~Allocated joint product costs~~	176	24
Cost of further processing	50	20
Sales value at the split-off point	140	40
Sales value after further processing	270	50

Sell or Process Further – Example

Should Sawmill, Inc. sell or process Lumber and/or Sawdust further?

Analysis of Sell or Process Further

	Per Log	
	Lumber	Sawdust
Sales value after further processing	$ 270	$ 50
Sales value at the split-off point	140	40
Incremental revenue	130	10
Cost of further processing	50	20
Profit (loss) from further processing	$ 80	$ (10)

The lumber should be processed further.

The sawdust should be sold at the split-off point.

Quick Check 12.2

Meatman Butchery purchases cow's loin for $10 per kg.

Initial cleaning of the cow's loin is $3 per kg.

Meatman can sell the cleaned cow's loin to consumers for $16 per kg.

Alternatively, Meatman can further process the cow's loin into tenderloins, at $20 per kg of loin.

When processing, 15% of original weight of the cow's loin will lost as scrap.

Tenderloins can be sold for $50 per kg. Scrap meat has no value.

What is the net advantage if Meatman decides to process 10 kg cow's loin into tenderloin to sell?

a. Net advantage of $65

b. Net advantage of $95

c. Net advantage of $140

d. Net advantage of $195

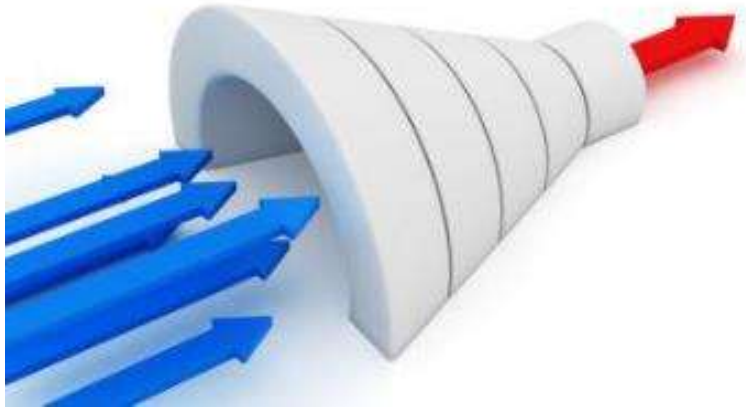
Learning Objective

Determine the most profitable use of a constrained resource and the value of obtaining more of the constrained resource.

Constrained Resource

When a limited resource of some type restricts the company's ability to satisfy demand, the company is said to have a **constraint**.





The machine or process that is limiting overall output is called the **bottleneck** – it is the constraint.

Utilization of a Constrained Resource

- Decide on best use of constrained resource
- Select product mix that can maximize profits with the added consideration of the constraint.
- Fixed costs remains unchanged within relevant range – IRRELAVANT. Hence, we focus on maximizing contribution margin.

Does it mean we should favor the product with the highest contribution margin?

Favor the product with the highest contribution margin per unit of constrained resource.

Illustration

- Time is a constrained resource.
- After classes, you have 6 hours a day to do tuition.

Company A – unlimited sessions	Company B – unlimited sessions
Each session takes 2 hrs of your time	Each session takes 3 hrs of your time
Pay per session $20	Pay per session $24

- **Should you source from Company B, since the pay per session is $24?**

Company A	Company B
$10 per hour	$8 per hour
3 sessions of 2 hrs each	2 sessions of 3 hrs each
Total: $60	Total: $48

Utilization of a Constrained Resource – Example

Ensign Company produces two products but has a constrained resource - Machine A1, being used at 100% of its capacity.

Machine A1 has a capacity of 2,400 minutes per week.

Should Ensign focus its efforts on Product 1 or Product 2?

	Product 1	Product 2
Selling price per unit	$ 60	$ 50
Less variable expenses per unit	36	35
Contribution margin per unit	$ 24	$ 15
Current demand per week (units)	2,000	2,200
Contribution margin ratio	40%	30%
Processing time required on machine A1 per unit	1.00 min.	0.50 min.

Utilization of a Constrained Resource – Example

Ensign Company produces two products but has a constrained resource - Machine A1, being used at 100% of its capacity.

Machine

Should Er

Do we just choose the product with the highest contribution margin?

	Product 1	Product 2
Selling price per unit	$ 60	$ 50
Less variable expenses per unit	36	35
Contribution margin per unit	$ 24	$ 15
Current demand per week (units)	2,000	2,200
Contribution margin ratio	40%	30%
Processing time required on machine A1 per unit	1.00 min.	0.50 min.

Utilization of a Constrained Resource – Example

1. What is the constrained resource?

2. How many units of each product can be processed per unit of that constrained resource?

3. What are the contribution margin per unit of that constrained resource?

Utilization of a Constrained Resource – Example

1. What is the constrained resource?

Machine A1

2. How many units of each product can be processed by a unit of that constrained resource?

	Product 1	**Product 2**
Unit of constrained resource	1 min	1 min
Processing time required per unit on A1	1 min	0.5 min
Number of units of product per unit of constrained resource	**1 unit per min**	**2 units per min**

3. What are the contribution margin per unit of that constrained resource?

	Product 1	**Product 2**
Contribution margin per unit	$24	$15
Number of units of product per min of A1	1 unit per min	2 units per min
Contribution margin per min of A1	**$24 CM per min**	**$30 CM per min**

Utilization of a Constrained Resource – Example

Ensign can maximize its contribution margin by first producing Product 2 to meet customer demand and then using any remaining capacity to produce Product 1.

Machine A1 has a capacity of 2,400 minutes per week.

Alloting Our Constrained Resource (Machine A1)

Weekly demand for Product 2		2,200	units
Time required per unit	×	0.50	min.
Total time required to make Product 2		1,100	min.
Total time available		2,400	min.
Time used to make Product 2		1,100	min.
Time available for Product 1		1,300	min.
Time required per unit	÷	1.00	min.
Production of Product 1		1,300	units

Utilization of a Constrained Resource – Example

According to the plan, we will produce 2,200 units of Product 2 and 1,300 of Product 1. Our contribution margin looks like this.

	Product 1	Product 2
Production and sales (units)	1,300	2,200
Contribution margin per unit	$ 24	$ 15
Total contribution margin	$ 31,200	$ 33,000

The total contribution margin for Ensign is $64,200.

Utilization of a Constrained Resource – Example

What if we first produce Product 1, and then use the remaining capacity to produce Product 2??

Priority is Product 2

	Product 1	Product 2	
Production and sales (units)	1,300	2,200	
Contribution margin per unit	$ 24	$ 15	
Total contribution margin	$ 31,200	$ 33,000	$64,100

Priority is Product 1

	Product 1	Product 2	
Production and sales (units)	2,000	800	
Contribution margin per unit	$ 24	$ 15	
Total contribution margin	$ 48,000	$ 12,000	$60,000

Contribution margin will be $4,100 lower.

Quick Check 12.3

Chip manufacturer, EEM is facing a bottleneck in its production process. It is doing an investigation so that it can better cater its product mix to maximize profits.

	Product A	Product B
Product units to produce	1,000	2,000
Part ABC (unit per pdt unit)	5	4
Part XYZ (unit per pdt unit)	7	7

In Inventory, there are 15,000 units of Part ABC and 20,000 units of Part XYZ.

Which is the constrained resource?

a. Part ABC

b. Part XYZ

c. Product B

d. No constrained resource

Managing Constraints

It is often possible for a manager to increase the capacity of a bottleneck, which is called relaxing (or elevating) the constraint, in numerous ways such as:

- Working overtime on the bottleneck.
- Subcontracting some of the processing that would be done at the bottleneck.
- Investing in additional machines at the bottleneck.
- Shifting workers from non-bottleneck processes to the bottleneck.
- Focusing business process improvement efforts on the bottleneck.
- Reducing defective units processed through the bottleneck.

These methods and ideas are all consistent with the Theory of Constraints – Managing constraints

End of topic